

VCU College of Engineering

CS 27-336 On-Device LLMs Real-Time NPC Interaction in Games

Project Proposal

Prepared for

Robert Iannello

ARM Sponsored Capstone Project

By

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Under the supervision of

Tom Arodz

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Executive Summary

This project, sponsored by Arm, focuses on improving how players interact with non-player characters (NPCs) in video games by using large language models (LLMs). Traditionally, NPCs rely on scripted dialogue and predetermined behaviors, which can make interactions feel repetitive and unrealistic. Our team's goal is to develop a system that allows NPCs to respond more naturally to players and react to different situations within the game. By using LLMs, we hope to demonstrate how artificial intelligence can create a more interactive and engaging gaming experience.

One of the main challenges of this project is running an LLM locally while maintaining acceptable game performance. LLMs require significant computing resources, and generating responses can introduce delays that affect gameplay. Our system will aim to maintain a stable 30–60 frames per second (FPS) while providing NPC responses within a reasonable amount of time. The project will initially focus on creating a working prototype for a PC or laptop, with mobile deployment as an end goal. We will also explore how an LLM can influence NPC behavior beyond basic conversations.

The project will involve researching available LLMs, evaluating development tools, and integrating the selected technology into a modern game engine. Throughout development, the team will test the system's performance, responsiveness, and ability to support dynamic NPC interactions. The final technical deliverables will include a playable prototype, source code, documentation, reproducible build instructions, and a short demonstration video.

Development will take place over two semesters, beginning with the project proposal in October 2026 and continuing through the final report in April 2027. Major milestones include a preliminary design report in December 2026 and a final project demonstration during the spring semester. By the end of the project, our team hopes to show that locally operated LLMs can be a practical way to improve NPC interactions without relying on cloud-based services. The results will help demonstrate the potential of on-device AI in future video game development.

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Section A. Problem Statement

Most modern video games continue to hand-author non-player characters (NPCs) through the use of behavior trees, state machines, and structured dialogue. Players often fail to connect with the rigid structure of NPCs due to static interaction. By utilizing large language models (LLMs), interaction with NPCs can adapt to the environment it is within. Dialogue no longer has to feel like blank words on a screen, and rather a flowing conversation. Even with recent use of LLM-based dialogue systems, the model generally acts as a chat interface over a static character. This limits NPCs to predictable behaviors, and fails to explore the possibilities of adaptation, reasoning, and coordination created by language models.

Several problems arise by utilizing LLMs to drive NPC behavior at runtime. Inference is computationally expensive and slow. Model output is freeform text that would need to be translated to influence in-game actions. NPCs will need constant awareness of changes in game state within a limited context. This all must run locally without cloud dependence, and eventually reach mobile deployment. Despite these challenges, companies like Arm strive to highlight the power of computing. In an ever evolving technological field the feasibility of advancement increases rapidly, this system will display such.

This project will design and implement an LLM-driven NPC mechanism that goes beyond conventional dialogue. The system will run locally at a stable 30-60 FPS to ensure playability and responsiveness. The result of this project will be a playable prototype with source code, reproducible build instructions, supporting documentation, and a short demo video.

Section B. Engineering Design Requirements

The design requirements for this project are given by ARM's goals and objectives. This project aims to propose and implement novel methods of using LLMs to control or interact with NPCs during runtime. Beyond dialogue systems, the focus is on new, creative mechanisms that showcase the potential of LLM-driven NPCs in real-time gameplay, that aims to run locally at a stable 30-60 FPS to ensure both playability and responsiveness.

B.1 Project Goals (i.e. Client Needs)

The project goals listed below are derived by ARM's goals and objectives. The focus here is to create new and creative mechanisms that showcase the potential of LLM-driven NPCs in real-time gameplay at runs at 30-60 FPS. This is to be run locally, whilst overall, achieving mobile deployment as the ultimate goal to meet.

1) To produce an NPC system at real-time using LLMs.

2) To run this NPC system at 30-60 FPS locally, ensuring that it runs smoothly and is responsive.

3) To demonstrate the potential of LLMs for real-time NPC behavior.

4) To create a dynamic and engaging NPC environment for players to experience and interact with.

5) To produce a functional prototype that demonstrates the project's concept and idea.

B.2 Design Objectives

The design objectives for the design are given by ARM's need for real-time LLMs for NPCs in games that can run in acceptable performance. These objectives will focus on the functionality, practicality, and performance of the planned design. The goals are as follows:

1) The design will allow LLM-driven NPCs to interact with the player.
2) The NPCs can respond based on different gameplay situations.
3) This design will work locally instead of it running over cloud services.
4) This design will run smoothly and be responsive.
5) This design will be run on a modern game engine (e.g., Unity, Unreal Engine, Godot)

B.3 Design Specifications and Constraints

The design specifications and constraints ensure that the LLM system for NPCs are within the respectable requirements and limitations, whilst also being operable. Here are the specifications and constraints:

Constraint	Type	Target/Description
NPC must be powered by LLM for interactions	**Functional**	**The player can respond and talk to NPCs, dynamically, during runtime**
Maintain acceptable game performance	**Functional**	**Must maintain around 30-60 FPS**
System must run locally	**Hardware**	**Ensure it runs on a LLM that is not cloud-based**
NPC responses must occur within an acceptable time	**Performance**	**Should not be at a time that seems unrealistic or unnatural (ex. anything more than 5-6+ seconds)**
System must operate on available hardware	**Hardware**	**Hardware that is capable of running LLM inference locally, PC and mobile**
Working prototype must work on selected game engine	**Software**	**Works on a modern game engine (e.g., Unity, Unreal Engine, Godot)**
Prototype must run on PC/laptop	**Hardware**	**PC/laptop must achieve acceptable performance when running the NPC-LLM system.**
Must show the potential of LLM-driven NPCs in real time gameplay	**Functional**	**Prototype must demonstrate LLM-driven NPC behaviour during real-time gameplay**
Project must be reproducible	**Usability**	**Must provide reproducible build instructions**
Design must be completed within capstone schedule	**Schedule**	**Must be completed within two semesters**

B.4 Codes and Standards

As of now, there are no legal codes that are directly applicable to this proof of concept of the use of LLM-driven NPC interaction in games. Regardless, this design will still consider any legal codes and requirements related to LLMs in general as we continue to develop to make sure that the design is responsible and reliable.

Section C. Scope of Work

The goal of the On-device LLMs for Real-time NPC interactions project is to create a more interesting, interactive user experience in games where interactions with NPCs feel more like a natural conversation rather than a preprogrammed response. Furthermore, ARM aims to showcase the power of LLMs and to prove this is no longer a theoretical application.

C.1 Deliverables

Technical Deliverables

- Video demo of prototype
- Source code
- Documentation of project design

Academic Deliverables

- Team Contract (9/4/2026)
- Preliminary Proposal (10/9/2026)
- Design Poster (11/13/2026)
- Preliminary Report (12/7/2026)
- Expo Abstract and Expo Design Poster (3/26/2027)
- Final Report (4/30/2027)

Potential Risks and Mitigation

- Third-Party Dependencies: There is a need to use open-source LLMs. To mitigate incompatibility risk, the team will stay consistent and lock in a single LLM.
- Technical complexity: The team will review each other's work to ensure quality and avoid confusion.

C.2 Milestones

Milestone	Description	Target Date
Team Contract	Rules of expectations for the team.	9/4/2026
Preliminary Proposal	Problem statement, design requirements, and project scope.	10/9/2026
Design Poster	Poster presenting the project	11/13/2026
Preliminary Report	Concept generation and evaluation, Design methodology, results and social impacts	12/7/2026
Expo Abstract	Finalize project abstract to present at expo	3/26/2027
Expo Design Poster	Finalize poster design to present at expo	3/26/2027
Final Report	Finalized project report.	4/30/2027

C.3 Resources

Software and Development Resources

- Game Engine (e.g., Unity, Unreal Engine, Godot)
- Game scripting language (e.g., Python, C++)
- GitHub

Hardware

- PC capable of running an LLM

Human Resources

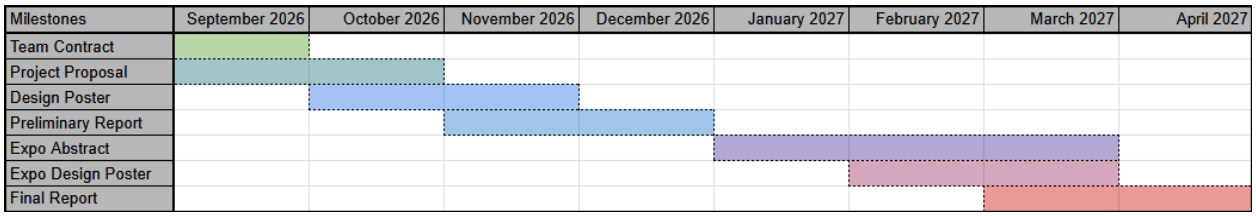
- Four senior Computer Science students are responsible for research, development, and testing.
- Faculty advisor
- Capital One sponsor representative.

Budget and Funding

- Expected budget requirement: NA

Appendix 1: Project Timeline

Provide a Gantt chart of similarly composed visual timeline showing the start and end dates of all completed tasks and how they are grouped together, overlapped, and linked together. Include all senior design requirements including design reports and Expo materials (i.e. Abstract, Poster, and Presentation). All major milestones should be included in the timeline.



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Team Member Name	*Strengths each member bring to the group*	*Other Info*	*Contact Info*
Wyatt Byrd	Report writing Java, Python		byrdwt@vcu.edu 7035822790
Benjamin Ruiz	Java Python C		ruizb@vcu.edu 7035877033
Jaskaran Singh	Experience with Java, Python, and C		singhj19@vcu.edu 8043127037
Caleb Rollston	Communication, Testing, Reporting		rollstonc@vcu.edu 804-839-9465

Other Stakeholders	*Notes*	*Contact Info*
Robert Iannello	ARM Sponsor	robert.iannello@arm.com
Tom Arodz	Faculty Advisor	tarodz@vcu.edu

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Culture Goals	***Actions***	***Warning Signs***
Work Ethic	- Diligently complete tasks in a timeful manner - Work efficiently as to meet deadlines and expectations of teammates, advisor, and sponsor	Missing due dates, incomplete tasks, underperforming
Respectful	- Being kind to others - Help one another	- When people talk over one another or don't give everyone a chance to contribute. - When people don't help one another and those people get left behind.
Communication	- Communicate with the team if there are any type of conflicts - Express any troubles or concerns - Keep in touch with the team	- Barely reaching out to the team - Not expressing any ideas

Step 3: Time Commitments, Meeting Structure, and Communication

Meeting Participants	***Frequency*** ***Dates and Times / Locations***	***Meeting Goals*** ***Responsible Party***
Students only	online - Wednesday 5 - bi-weekly	Define tasks, ensure team is on track
Students + Advisor	Online - Thursday 2:30 - bi-weekly (next meeting 9/17)	
Students + Sponsor	Still awaiting response	

Step 4: Determine Individual Roles and Responsibilities

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.
6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

Team Member	***Role(s)***	***Responsibilities***
Caleb Rollston	Testing Engineer	Testing the product and validating
Benjamin Ruiz	Financial manager	Monitors and calculate teams budget and spending.
Wyatt Byrd	Project Manager/Logistics Manager	Management of communication between groupmates, sponsor, and advisor. Documentation of meetings, agenda, tasks. Assuring deadlines are met.
Jaskaran Singh	Systems Engineer/Manufacturing Engineer	Manages the system architecture of the project and the designs for the project and ensures that it meets all requirements.

Step 5: Agree to the above team contract

Team Member:Caleb Rollston *Signature:* 



Team Member:Benjamin Ruiz *Signature:*

Team Member:Wyatt Byrd *Signature:* 

Team Member: Jaskaran Singh *Signature:* 

References

Provide a numbered list of all references in order of appearance using APA citation format. The reference page should begin on a new page as shown here.

[1] VCU Writing Center. (2021, September 8). *APA Citation: A guide to formatting in APA style.* Retrieved September 2, 2024. https://writing.vcu.edu/student-resources/apa-citations/
[2] Teach Engineering. *Engineering Design Process*. TeachEngineering.org. Retreived September 2, 2024. https://www.teachengineering.org/populartopics/designprocess